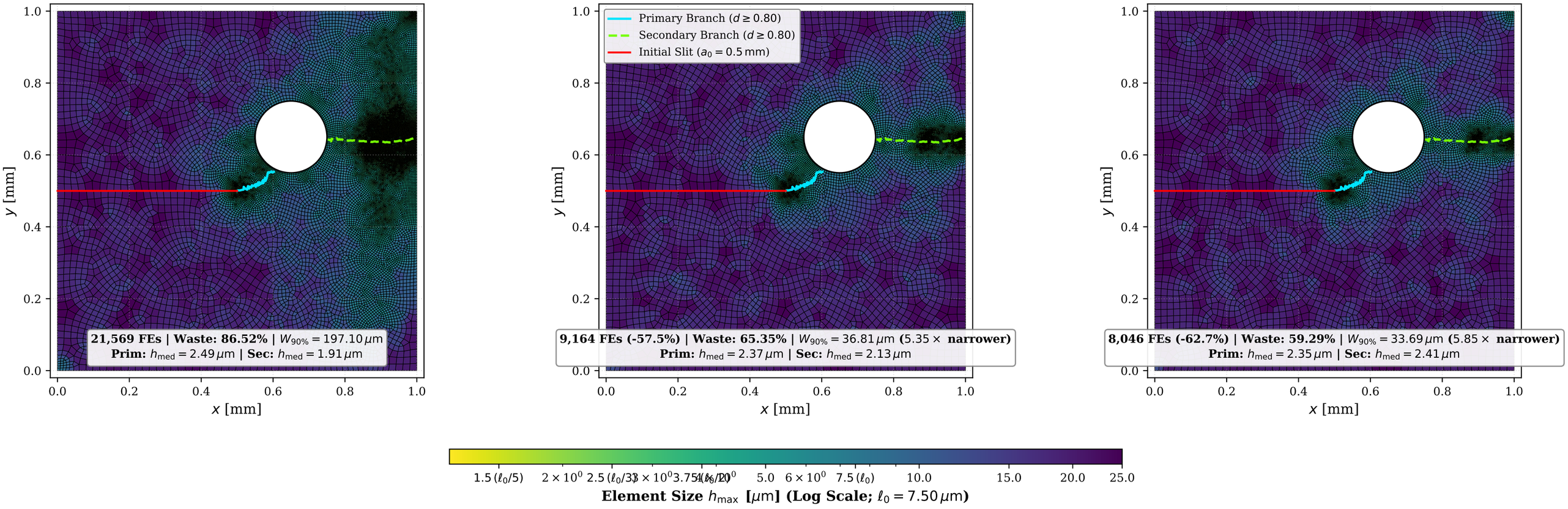

(a) Combined Baseline ($e_{tgt}^{prim} = 5\%, e_{tgt}^{sec} = 5\%$)
(21,569 elements, Broad Diffuse Refinement)
(b) Selected Candidate E10 ($e_{tgt}^{prim} = 5\%, e_{tgt}^{sec} = 10\%$)
(9,164 elements, Narrow Crack-Following)
(c) Selected Candidate E12 ($e_{tgt}^{prim} = 5\%, e_{tgt}^{sec} = 12\%$)
(8,046 elements, Lean Crack-Following)
Primary Branch ($d \geq 0.80$)
Secondary Branch ($d \geq 0.80$)
Initial Slit ($a_0 = 0.5$ mm)
21,569 FEs | Waste: 86.52% | $W_{90\%} = 197.10\,\mu$m
Prim: $h_{med} = 2.49\,\mu$m | Sec: $h_{med} = 1.91\,\mu$m
9,164 FEs (-57.5%) | Waste: 65.35% | $W_{90\%} = 36.81\,\mu$m (5.35 × narrower)
Prim: $h_{med} = 2.37\,\mu$m | Sec: $h_{med} = 2.13\,\mu$m
8,046 FEs (-62.7%) | Waste: 59.29% | $W_{90\%} = 33.69\,\mu$m (5.85 × narrower)
Prim: $h_{med} = 2.35\,\mu$m | Sec: $h_{med} = 2.41\,\mu$m
x [mm]
y [mm]
0.0
0.2
0.4
0.6
0.8
1.0
1.5 ($\ell_0/5$)
2 × 10^0
2.5 ($\ell_0/3$)
3 × 10^0
3.75 ($\ell_0/2$)
5.0
6 × 10^0
7.5 (ℓ_0)
10.0
15.0
20.0
25.0
Element Size h_{max} [μm] (Log Scale; $\ell_0 = 7.50\,\mu$m)